

# Gregory Ling

gling@ctrl.com | ctrl.com | github.com/glingy

## Education

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### Iowa State University (ISU) – Ames, IA

- B.S. Computer Engineering – GPA: 3.99
- M.S. with Thesis in Computer Engineering – GPA: 4.00

Fall 2020 – Fall 2023  
Fall 2023 – Fall 2024

## Work Experience

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### Firmware Engineer IBM – Rochester, MN

2024 – Present

- C++ development of the Power Hypervisor for IBM's Power mainframes, focused on codebase modernization
- Implemented additional capturing of stack traces on failure to aid in debugging without access to a system
- Wrote VSCode extension to aid in development and debug process

### Teaching Assistant, Iowa State University – Ames, IA

Fall 2023 – Fall 2024

- TA for Intro to Digital Logic, Advanced Embedded Systems, and Neural Network Microarchitecture Design courses

### Software Engineer Intern, Altec – St. Joseph, MO

Summer 2023

- Yocto Linux development environment using Postgresql and MQTT on an i.MX6 processor
- Integrated zlib in C++ to compress traces being sent, and reduced kernel boot time and size footprint

### Research Assistant (REU), Iowa State University – Ames, IA

Summer 2022 – Spring 2023

- Extremely low power system consisting of an FPGA and MSP430 microcontroller running from a wireless harvester.
- Designed a very low-power, intermittently-powered neural network accelerator in VHDL
- Custom 16-wire SPI protocol by using the DMA to read/write to the PORT register of the MSP430
- Additionally wrote firmware and tested a harness of 5 MSP430 MCUs to provide physical validation of simulation results.
- Published in <https://dl.acm.org/doi/10.1016/j.peva.2024.102440>

## Project Highlights

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### Solar Car Club at ISU (American Solar Challenge PrISum #9)

- Software Lead for three years, building the team from 1 to about 15 people
- Designed custom development board with embedded debugger to provide new members with a cost-effective playground to experiment while learning (see <https://ctrl.com/projects/playground> for more details)
- Migrated to new IDE and build system, focused on direct peripheral management with emphasis on reading the datasheet
- Designed custom bootloader to allow remote flashing over the car's CAN bus through the telemetry system
- Taught new members to understand a datasheet and use the full capabilities of the ATSAMC21 processor
- Led the telemetry project: an embedded Linux system built using Yocto, running a lightweight C++ web server with Vue.js frontend to transmit the car's status over WiFi or LTE in real time

### ASIC Design Club (ChipForge)

- Master's thesis project was to create a student organization centered around designing ASICs to submit for fabrication
- Two ASICs have been returned from fabrication and tested by the members and senior design students

### CPRE 488 Open-Ended Design Project (Sparky)

- Integrated a Roomba with a Digilent ZedBoard, Raspberry Pi, webcam, ESP32, bluetooth serial module and game controller, foam dart turret, and CrazyFlie tracker
- End result could be remotely controlled via a wireless game controller and Quest VR headset
- Turret could fire foam darts, or an autonomous drone would fly to provide target practice and land on a simulated "hit"

### Handheld Game Console

- Designed a custom handheld game console using an ATSAMC21 (Adafruit Adalogger M0) with an LCD, SD card reader, joystick, 3D-printed case, and custom USB & SD card bootloader
- Used only CMSIS headers and the datasheet with no libraries

### NXT/EV3 Infrared Remote Control "Sensor"

- Built an infrared remote that connects to a LEGO NXT/EV3 as a sensor, controlling LEGO Power Functions from the EV3.
- Designed the PCB using a TI MSP430 and wrote firmware in C using PlatformIO

## Skills

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**Processors:** TM4C, ATSAMx21, ATMEGA, MSP430, STM32, i.MX6, POWER

**Languages:** C, C++, ASM (8-bit AVR, 32-bit ARM, 32-bit MIPS, POWER), Verilog, VHDL, Swift, JavaScript (Browser & Node.js), TypeScript, CoffeeScript, Vue.js, HTML, PHP, CSS, Java, Python, UNIX

**Tools:** Git, Agile Development, Laser Cutting, 3D Printing (FreeCAD, OpenSCAD, Fusion 360), PCB Design (Eagle, KiCad, Altium), HDL (Vivado, ModelSim)